

Installation Guide

Environment Setup & Virtualenv Configurations

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Installation Guide

This document describes how to install the required environment, configure settings, and prepare the database and model binaries for CreditGuard AI.

1. Prerequisites

Before installing, ensure your machine has the following tools:

- Python 3.10 or 3.11 (recommended version is 3.10).
 - Git (for version control).
 - Docker Desktop (optional, for containerized execution).
 - Pip (Python package installer).
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2. Step-by-Step Installation

Step 2.1: Clone the Repository

Clone this repository to your local system and navigate to the root directory:

```
git clone https://github.com/skmdshariff143-ai/Credit-Card-Approval-Prediction.git
cd Credit-Card-Approval-Prediction
```

Step 2.2: Setup Virtual Environment

It is highly recommended to use a virtual environment to prevent package version conflicts:

- Windows (PowerShell):

```
python -m venv venv
.\venv\Scripts\Activate.ps1
```

- macOS / Linux:

```
python3 -m venv venv
source venv/bin/activate
```

Step 2.3: Install Package Dependencies

Install the required packages listed in `requirements.txt`:

```
pip install --upgrade pip
pip install -r requirements.txt
pip install -e .
```

Step 2.4: Configure Environment Settings

Create a `.env` file in the root directory by copying the sample config:

```
copy .env.example .env
```

Open ``.env`` and verify the settings:

```
FLASK_ENV=development  
SECRET_KEY=dev_secret_key_12345  
LOG_LEVEL=INFO
```

Step 2.5: Train Model Binaries

Run the pipeline script to ingest raw data, preprocess variables, train models, and serialize the best candidate model:

```
python src/main.py
```

Verify that the output binaries are created:

- `models/best_model.pkl``
- `models/scaler.pkl``
- `models/encoder.pkl``
- `models/preprocessing_pipeline.pkl``

3. Running the Server

Start the local Flask development server:

```
python app/app.py
```

Open your browser and navigate to ``http://localhost:5000`` to access the portal!